

UK Maths Trust

SENIOR MATHEMATICAL CHALLENGE

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MARKETS

For reasons of space, these solutions are necessarily brief.

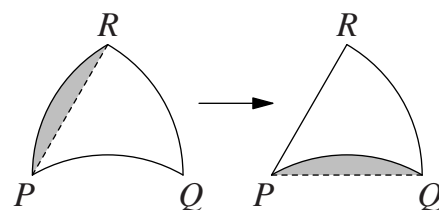
More in-depth, extended solutions, including exercises for further investigation, are available on the UKMT website.

A version of this document including each of the questions alongside its solution is also available on the UKMT website:

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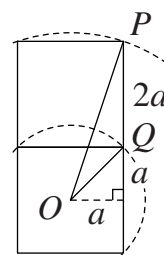
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1. **C** Pablo can use 1, 8, 27 or 64 small cubes to build solid cubes. As he builds the largest solid cube possible, he has $100 - 64 = 36$ small cubes left over.
2. **E** The largest equilateral triangle is made from 9 small equilateral triangles. The two shaded triangular areas can be rearranged to form a parallelogram made from four small equilateral triangles. Therefore, the shaded area is $\frac{4}{9}$ of the large triangle.
3. **E** Rewriting $25^2 - 24^2 - 23^2 + 22^2$ as $25^2 - 24^2 - (23^2 - 22^2)$ and using the difference of two squares twice gives $(25 + 24)(25 - 24) - (23 + 22)(23 - 22) = 49 \times 1 - 45 \times 1 = 4$.
4. **A** The sum of Sonia's three numbers, 46, is even. Therefore the prime and square are either both even or both odd. As all 2-digit primes are odd, the square must also be odd. There is exactly one odd, 2-digit square smaller than 46, which is 25, so 25 is Sonia's square. The sum of the remaining prime and even number is $46 - 25 = 21$. There is one possible pair of 2-digit integers satisfying prime + even = 21 which is $11 + 10 = 21$. Therefore Sonia's even number is 10.
5. **A** The area of the largest circle is $\pi \times 3^2 = 9\pi$. The shaded area is $\pi \times 2^2 - \pi \times 1^2 = 4\pi - \pi = 3\pi$. Therefore the ratio of the shaded area to the area of the largest circle is $3\pi : 9\pi = 1 : 3$.
6. **B** In order to solve the equation, the following process is repeated three times: subtract 1 from each side then square. This produces the equation $x = (((3 - 1)^2 - 1)^2 - 1)^2 = 64$.
7. **D** The sequence can be viewed as alternating between two linear sub-sequences, 1, 7, 13, 19, ... and 5, 11, 17, 23, Hence the $(2n - 1)$ th term of the original sequence is of the form $(6n - 5)$ and the $(2n)$ th term is of the form $(6n - 1)$. The 100th term of the original sequence is the 50th term of the second sub-sequence, $6n - 1 = 6 \times 50 - 1 = 299$.
8. **B** By drawing chords PR and PQ and moving the region which lies between chord PR and arc PR to fill the space of the same size between chord PQ and arc PQ we form a sector of radius 1. The angle RPQ is 60° . Therefore the area of the shape is $\frac{1}{6} \times \pi \times 1^2 = \frac{\pi}{6}$.



9. C Labelling the squares in the line 1 to 50, we start by shading squares 1 and 50. One possible maximal shading is all of the odd numbered squares 1 to 47 inclusive, but not 49 as 49 and 50 are adjacent and so can't both be shaded. This is 25 shaded squares. One possible minimal shading using the largest permitted gaps is every fourth square from 1 to 45 inclusive namely 1, 5, 9, ..., 41, 45. As before, 49 can't be shaded but either 47 or 48 must be. This is 14 shaded squares. So the difference is $25 - 14 = 11$ shaded squares.
10. E The prime factorisation of 2025 is $3 \times 3 \times 3 \times 3 \times 5 \times 5$. Therefore squares which are factors of 2025 are 1, $3^2 = 9$, $5^2 = 25$, $(3^2)^2 = 81$, $(3 \times 5)^2 = 225$, $(3^2 \times 5)^2 = 2025$, so 6 in total.
11. B Triangles PQR and PNM are both right-angled and contain $\angle MPN$. The triangles are therefore similar, with shortest sides QR and MN respectively. Using Pythagoras' Theorem, $QR = \sqrt{25^2 - 20^2} = 15$. For triangle PNM to have area which is half that of triangle PQR , the length scale factor must be $\sqrt{\frac{1}{2}} = \frac{\sqrt{2}}{2}$. The length of MN is then $15 \times \frac{\sqrt{2}}{2} = \frac{15}{2}\sqrt{2}$.
12. D A 3-digit number is divisible by 3 if, and only if, the sum of its digits is divisible by 3. The digits of Ayain's numbers are three different numbers from the set 1, 3, 5, 7, 9. Hence each number is obtained from one of the sets (1, 3, 5), (1, 5, 9), (3, 5, 7), or (5, 7, 9). For each possible triple, there are 6 possible orders, e.g. 135, 153, 315, 351, 513, 531. So in all, $4 \times 6 = 24$ of Ayain's numbers are divisible by 3.

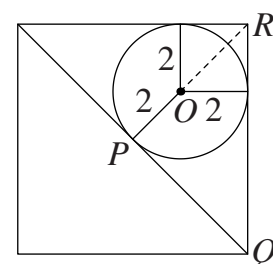
13. D Let O be the centre of the middle square, Q be a vertex of the middle square and P be a vertex of the square above as shown. Therefore OQ is a radius of the smaller circle and OP is a radius of the larger circle. Using Pythagoras' Theorem twice gives $(OQ)^2 = a^2 + a^2 = 2a^2$ and $(OP)^2 = a^2 + (3a)^2 = 10a^2$. The shaded area is therefore $\pi \times (OP)^2 - \pi \times (OQ)^2 = \pi(10a^2 - 2a^2) = 8\pi a^2$.



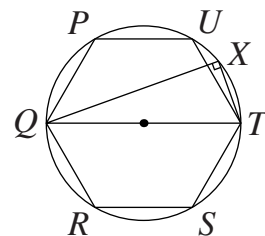
14. B First rearrange $x + \frac{1}{y} = 2$ (1) and $y + \frac{1}{x} = \frac{9}{4}$ (2) to $xy + 1 = 2y$ (1') and $xy + 1 = \frac{9x}{4}$ (2'). Equating the RHS of (1') and (2'), gives $2y = \frac{9x}{4}$ and therefore $y = \frac{9x}{8}$. Substituting into (1'), gives $x \times \frac{9x}{8} + 1 = 2 \times \frac{9x}{8}$ which simplifies to $9x^2 - 18x + 8 = 0$ with solutions $x = \frac{4}{3}$ and $x = \frac{2}{3}$. The difference between the values of x is $\frac{4}{3} - \frac{2}{3} = \frac{2}{3}$.

15. B We start by writing expressions for P and Q as $P = n - 1$ and $Q = 99 - n$. Note that if $n = 50$, then both P and Q are 49 and so $R = 0$, which is not prime. Now consider values of n for which $1 < n < 50$. In this case, $Q > P$ therefore $R = Q - P = (99 - n) - (n - 1) = 100 - 2n$. In order for this value of R to be prime, $100 - 2n = 2$ and so $n = 49$. However, considering values of n for which $50 < n < 99$ leads to $P > Q$. In this case $R = P - Q = (n - 1) - (99 - n) = 2n - 100$. In order for this value of R to be prime, $2n - 100 = 2$ and therefore $n = 51$. Hence there are two possible values of n which lead to R being prime, $n = 49$ and $n = 51$.

16. A Let the centre of the circle be O , the centre of the square be P and vertices of the square be Q and R as shown. The radius of the circle is 2, so $PR = PO + OR = 2 + 2\sqrt{2}$. As triangle RPQ is right-angled and isosceles its sides are in the ratio $1 : 1 : \sqrt{2}$. The length of the side of the square, QR , is therefore $\sqrt{2}(2 + 2\sqrt{2}) = 2\sqrt{2} + 4 = 4 + 2\sqrt{2}$.



17. **D** We note that there are 21 possible sums: 1 sum of all five numbers, 5 sums of four numbers, 5 sums of three adjacent numbers, 5 sums of two adjacent numbers and 5 single numbers. For brevity and without loss we will consider in this last case that a single number is a sum of 1 number. As there are 21 possible sums and we seek to create each total from 1 to 21 inclusive, every sum must occur exactly once; there can be no repeated sums. Given that 1 and 5 are already shown, the remaining numbers must sum to $21 - (1 + 5) = 15$. To create a sum of 2, we must use a 2, as $1 + 1$ would be a repeat. The remaining two integers must have a total of $21 - (1 + 5 + 2) = 13$. The only possible pairs are 6, 7 or 4, 9 or 3, 10. We first discount using a 6 because the adjacent 1 and 5 already have a total of 6. We next discount 4, 9. In this case, as there is no single 3, the 2 would be placed next to the 1 to create a sum of 3. Placing the 4 next to the 5 would create 9 in two ways and placing the 4 next to the 2 would create 6 in two ways. So now we have our five integers, 1, 2, 3, 5 and 10. To create a sum of 4, the 3 must be next to the 1. All the totals from 1 to 6 inclusive are now possible. To create a sum of 7, the 2 must be next to the 5, leaving $x = 10$. All sums from 1 to 21 inclusive are now possible, which is left as an exercise for the reader.
18. **C** Listing the 2-digit multiples of 17 and of 23 gives us the numbers which we can use when building the integer N , namely 17, 34, 51, 68, 85 and 23, 46, 69, 92. The first digit is 3, so using 34, the second digit is 4 and using 46, the third digit is 6. When we reach 6, there are two choices, 68 or 69. Choosing 8, the following three digits would be 5, 1, 7. However as there are no multiples of either 17 or 23 in the 70s we reach an end. This can only happen at the very end of N . If however we choose 69, then a cycle of five digits is formed, ..346923.. . Our positive integer N is either formed from 405 cycles of 34692 (ending in 2) or from 404 cycles of 34692 followed by 34685 (ending in 5). The value of $p + q$ is therefore $2 + 5 = 7$.
19. **C** Let the cost of a shirt be x , the cost of a pair of shorts be y and the cost of a pair of socks be z . A 'complete' kit then costs $x + y + z$. Eden has $9x$ so we want to write $9x$ in the form $k(x + y + z)$ for some integer k , $1 \leq k \leq 8$. We are given that $2x = 3y + z$ and $5x = 7y + 4z$. We need a linear combination of these two equations such that in the resulting equation, the coefficients of y and z are equal. By inspection, subtracting three times the first equation from twice the second shows that $4x = 5y + 5z$. Then $9x = 5x + 4x = 5x + 5y + 5z = 5(x + y + z)$ and so five complete football kits could be bought for the same amount of money as nine shirts.
20. **E** Starting with $\frac{x + 2026xy - y}{2y - 2025xy - 2x}$ and dividing both the numerator and denominator of the expression by xy gives $\frac{\frac{1}{y} + 2026 - \frac{1}{x}}{\frac{2}{x} - 2025 - \frac{2}{y}} = \frac{-\left(\frac{1}{x} - \frac{1}{y}\right) + 2026}{2\left(\frac{1}{x} - \frac{1}{y}\right) - 2025} = \frac{-2025 + 2026}{2 \times 2025 - 2025} = \frac{1}{2025}$.
21. **D** Placing X in a general position on the circle and creating a triangle by joining X to points at opposite ends of a diameter, for example Q and T as shown, we can use 'the angle in a semicircle is 90° '. As the angle at X is 90° we can use Pythagoras' Theorem, $XQ^2 + XT^2 = QT^2 = 10^2 = 100$. Similarly, with RU as diameter $XR^2 + XU^2 = RU^2 = 100$ and with SP as diameter $XS^2 + XP^2 = SP^2 = 100$.



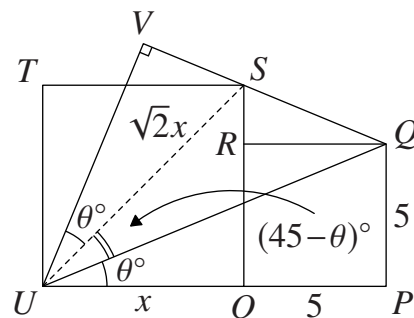
Therefore $XP^2 + XQ^2 + XR^2 + XS^2 + XT^2 + XU^2 = 3 \times 100 = 300$. Note that if X is placed at a vertex, there would only be five chords but the total would be the same.

22. B There are $(n + 12)$ animals. The number of ants that each animal eats is $\frac{(n^2 + 20n + 25)}{(n + 12)} = \frac{(n + 12)(n + 8) - 71}{(n + 12)} = (n + 8) - \frac{71}{(n + 12)}$. For the number of ants eaten by each animal to be a whole number, $(n + 12)$ must be a factor of 71. As 71 is prime, the only possibility is that $n + 12 = 71$ therefore $n = 59$. The number of ants eaten by each animal is then $59 + 8 - 1 = 66$.

23. C Let f be the maximum mark in each test. We can write two expressions for f . Firstly, f is the difference in Jemima's total score between taking $(n - 1)$ tests and taking $(n - 2)$ tests. So $f = (m + 4)(n - 1) - m(n - 2) = mn + 4n - m - 4 - (mn - 2m) = 4n + m - 4$. Secondly, f is the difference in her total score between taking n tests and taking $(n - 1)$ tests. So $f = (m + 4 + 3)n - (m + 4)(n - 1) = mn + 7n - (mn + 4n - m - 4) = 3n + m + 4$. Equating these two expressions for f gives $4n + m - 4 = 3n + m + 4$ and therefore $n = 8$.

24. A First consider $0 < x < 1$. For all values of x in this interval, $F(x) = 0$ and therefore $y = x^{F(x)} = x^0 = 1$. This eliminates options B, C and D. Now consider $-2 < x < -1$. For all values of x in this interval, $F(x) = -2$. Therefore $y = x^{F(x)} = x^{-2} = \frac{1}{x^2} > 0$, as $x^2 > 0$. This eliminates option E. Using a similar approach, we can show that, for $-1 < x < 0$, $F(x) = \frac{1}{x}$ and, for $1 < x < 2$, $F(x) = x$ which are consistent with the graph in the remaining option A.

25. C Let $OU = x$ and $\angle QUP = \theta^\circ$. As $\angle SUO = 45^\circ$, $\angle SUQ = 45^\circ - \theta^\circ$. As $\angle VUQ = 45^\circ$, $\angle VUS = 45^\circ - (45 - \theta)^\circ = \theta^\circ$. Thus right-angled triangles QUP and VUS are similar. In each triangle, we have enough information to consider $\cos \theta$. In triangle QUP , the adjacent side $UP = x + 5$. Using Pythagoras' Theorem, the hypotenuse $QU = \sqrt{(x + 5)^2 + 5^2} = \sqrt{x^2 + 10x + 50}$. In triangle VUS , the adjacent side VU is a shorter side of the right-angled isosceles triangle VUQ , therefore $VU = \frac{1}{\sqrt{2}}QU = \frac{\sqrt{x^2 + 10x + 50}}{\sqrt{2}}$. The hypotenuse SU is also the diagonal of square $OSTU$ therefore $SU = \sqrt{2}x$.



This gives two expressions for $\cos \theta$, namely $\frac{x + 5}{\sqrt{x^2 + 10x + 50}}$ and $\frac{\frac{\sqrt{x^2 + 10x + 50}}{\sqrt{2}}}{\sqrt{2}x}$. Equating these and rearranging gives $2x(x + 5) = x^2 + 10x + 50$. Therefore $2x^2 + 10x = x^2 + 10x + 50$ and so $x^2 = 50$. Hence the area of square $OSTU$ is 50.